



# Write your own McStas component

- **Aim:**
- Write a component that models a semi-permeable mirror. i.e. a mirror with reflectivity < 1, which also uses focused sampling.
- **Hints:**
- Step by step instructions in README on github / Wednesday – item 4  
[https://github.com/mccode-dev/Schools/tree/master/2025/ISIS\\_November\\_2025/3\\_Thursday\\_November\\_6th/4\\_Writing\\_a\\_component/Write\\_a\\_component](https://github.com/mccode-dev/Schools/tree/master/2025/ISIS_November_2025/3_Thursday_November_6th/4_Writing_a_component/Write_a_component)
- Start with an **Arm** – it has most of the McStas structure present already.
- Focused sampling is governed by:

$$\pi_i = \frac{P}{f_{mc}}$$

$\pi_i$  : weight multiplier

$P$  : physical probability

$f_{mc}$  : Monte Carlo probability